

Review article

Quality assessment of software requirements using artificial intelligence methods: A systematic literature review

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ABSTRACT

Context: The quality of requirements specifications is a critical success factor in software development. Assuring high-quality requirements, specifically in an automated way, poses a significant challenge due to their unstructured and multi-modal character. With the rise of deep learning and large language models (LLMs), new opportunities have developed to assess the quality of requirements automatically, particularly user stories in the context of agile software engineering, where short development cycles require efficient tool support.

Objective: This study aims to systematically review and investigate the current landscape of approaches based on artificial intelligence techniques such as natural language processing and deep learning for assessing the quality of software requirements. The investigation focuses on the artificial intelligence techniques adopted, quality aspects considered, datasets used to tune and evaluate the proposed approaches, and their performance.

Method: We conducted a systematic literature review of 26 peer-reviewed papers published between 2019 and 2025. We selected the papers after a title and abstract review of 353 papers identified through a literature databases query and forward-backward snowballing.

Results: The results reveal significant overlap among considered quality aspects, which can be mapped onto the higher-order requirements quality model INVEST. Most studies focus on assessing requirement quality rather than improving requirements and rely heavily on synthetic and public datasets. LLMs have rapidly gained popularity since 2023, though model evaluation strategies remain inconsistent. Metrics such as accuracy, precision, recall, and F1-Score are common, yet a few studies use semantic or expert-based evaluations.

Conclusion: The field is evolving toward LLM-driven, semantically rich models, yet lacks methodological standardization, reproducible datasets for evaluating the models, and integration of the approaches with real-world requirements engineering processes. Future work should address these limitations by developing benchmark datasets, standardizing evaluation metrics, and exploring hybrid systems that combine AI-based and traditional requirements quality assurance approaches.

Contents

1. Introduction	2
2. Background and related work	3
2.1. Quality assurance of software requirements	3
2.2. AI for software requirements engineering	3
2.3. AI for requirements quality assessment	3
2.4. Objective of this review	4
3. Research questions	5
4. Review method	5
4.1. Data sources	5
4.2. Search strategy	5
4.2.1. Inclusion and exclusion criteria	6
4.2.2. Quality assessment	6

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4.2.3.	Snowballing strategy and results.....	7
4.3.	Demographic analysis.....	7
4.4.	Data extraction.....	7
5.	Results.....	9
5.1.	Studies overview.....	9
5.2.	RQ1: What quality aspects and associated metrics are considered to assess the quality of software requirements?.....	9
5.2.1.	Quality aspects: Origins and usage frequencies.....	9
5.2.2.	Quality aspects: Usage purposes and RE phases.....	11
5.2.3.	Quality metrics and quality measurement.....	11
5.3.	RQ2: What datasets were used to create, evaluate, and adjust requirements quality assessment models?.....	11
5.4.	RQ3: What AI methods and models are used for the automated assessment of the software requirements' quality?.....	12
5.4.1.	Overview of AI-based quality assessment methods.....	14
5.4.2.	Evaluation of the AI-based quality assessment methods.....	15
5.4.3.	Evolution AI-based quality assessment methods over time.....	16
6.	Discussion.....	16
6.1.	What dimensions of requirements quality are considered?.....	16
6.2.	For what purpose and in which RE phase is the quality of requirements assessed?.....	17
6.3.	What metrics are used to quantify the quality of requirements?.....	18
6.4.	What data is used to develop and validate AI-based methods for requirements quality assessment?.....	18
6.5.	What AI methods and models are adapted for assessing the quality of requirements?.....	18
6.6.	How the performance of AI-based quality assessment methods is evaluated?.....	19
6.7.	How quality assessment methods exploit advancements in AI technologies?.....	19
7.	Threats to validity.....	19
8.	Research opportunities and challenges.....	19
9.	Directions for future research.....	20
10.	Conclusions.....	20
	CRedit authorship contribution statement.....	20
	Declaration of competing interest.....	20
	Acknowledgments.....	20
	Data availability.....	20
	References.....	20

1. Introduction

Requirements engineering (RE) is critical for the success of software projects, ensuring that all stakeholders' needs are accurately captured and addressed. Compared to other phases of the software development life cycle (SDLC), ensuring quality during requirement specification is especially demanding due to the heavy reliance on human experts and the fact that the produced artifacts are predominantly textual and contain rich semantics that are often not directly machine-readable. Therefore, software process automation solutions, such as software measurement and data analysis, are difficult to apply to RE. With the rise of novel artificial intelligence (AI) techniques, in particular large language models (LLMs), and their application in various domains, there is an increasing interest in their potential to support RE, specifically for assessing the quality of software requirements. Characteristics such as automation and human-like interaction predispose LLMs for RE, especially in the context of agile software engineering, where short development cycles require efficient tools.

This systematic literature review (SLR) explores the current landscape of the AI applications for RE, specifically for assessing the quality of software requirements. Our goal is to investigate which of the AI techniques have been adopted for this purpose and how, as well as their performance in terms of the effectiveness of the assessments they provided.

Despite increasing research activity at the intersection of RE and AI, particularly in the context of natural language processing (NLP) and deep learning (DL), there is a lack of comprehensive and structured overviews of this field. Prior literature focuses primarily on using AI to support constructive activities of RE such as requirement elicitation [1], classification [2,3], and specification [4], yet largely overlooks assessing the actual quality of the requirements, particularly regarding evaluating custom-defined quality aspects and implementing targeted improvements for identified deficits.

Existing reviews tend to either address broader RE challenges, such

as the integration of AI across the full RE lifecycle [5,6], the lack of tool-supported ambiguity resolution in user stories [7], or the overall automation potential in RE tasks [8], or focus exclusively on traditional NLP methods without considering the implications of LLMs or the diversity of evaluation strategies [4]. To address this gap, we review and synthesize recent literature on the applications of AI for assessing software requirements.

In contrast to prior reviews that broadly examined AI in RE or that predate the emergence of LLMs, this study focuses explicitly on the intersection of AI and requirements quality assessment. Specifically, the review concentrates on studies that employ AI techniques to assess the quality of software requirements. This review therefore aims, for the first time, to provide a comprehensive synthesis that systematically maps the conceptual foundations of requirements quality assessment across AI-based studies, revealing whether a unifying taxonomy of quality aspects exists and how current research operationalizes them. Furthermore, the SLR aims to address the temporal gap left by earlier literature reviews that predate the rise of LLMs and Generative AI (GenAI) and therefore do not capture recent methodological advances enabling the evaluation of richer and more nuanced quality aspects.

By pursuing these objectives, this study enhances our understanding of (1) which quality aspects have been considered in current research and how they overlap or diverge across studies, (2) which datasets are used for developing and validating AI-based quality assessment models, and (3) which AI techniques are applied to perform these assessments.

The reminder of this article is organized as follows: In Section 2, we provide background knowledge and review related work. Section 3 specifies the Research Questions (RQs) and Section 4 describes the corresponding method used for this review. Section 5 presents the outcomes of the review, followed by their discussion in Section 6. Section 7 discusses potential threats to the validity of the review results. Finally, Section 10 concludes the article with a summary of implications on the RE practice and further work perspectives.

2. Background and related work

2.1. Quality assurance of software requirements

RE is the area of software systems' engineering, focusing on the discovery, development, tracing, analysis, qualification, communication, and management of requirements [9]. Identifying and specifying the needs and constraints of relevant stakeholders is the first phase of a software system development life cycle [10].

High-quality requirements are the foundation of successful software development projects. Clear and understandable requirements improve communication among stakeholders. They ensure that stakeholders' needs will be met, increasing their satisfaction, and provide the basis for user-needs-oriented testing. Requirements deficits may lead to expensive rework, especially when detected in late phases of the SDLC. From the project perspective, well-defined requirements allow for reliable planning and effective control of the project scope.

Various methods have been proposed over the years to ensure that requirements reflect true stakeholder intents and are of high quality. Basic strategies to ensure quality include *constructive* and *analytical* ones. Examples of *constructive* methods to "build quality in" during requirements development include the use of structured interviews and focus groups [11], the creation of prototypes and mock-ups [12], and the specification of requirements in a structured and formal manner [13]. The quality of the resulting requirements is then evaluated using *analytical* "test quality in" methods such as manual inspections and walkthroughs [14], automatic quality checks against predefined rules [15], e.g., avoidance of specific ambiguous words or phrases [16], traceability checks [17], formal modeling and testing [13].

Requirements analysis can furthermore be guided by a predefined requirements quality model, typically defined as a set of quality criteria. One example is the widely recognized INVEST model by Bill Wake [18]. It defines six core criteria (Independent, Negotiable, Valuable, Estimable, Small, and Testable) for evaluating the quality of agile requirements. Extensions such as the Improved INVEST model by Brockenbrough and Salinas [19] and the Quality User Story (QUS) framework by Lucassen et al. [20] have further refined these principles by introducing attributes such as structure, traceability, conceptual clarity, and uniformity. Other frameworks, including QuARS by Lami [21] and ALAS by Zhang et al. [22], offer alternative or complementary perspectives, emphasizing readability, ambiguity avoidance, or formal compliance.

Although many automatic tools have been proposed [23], RE remains a complex and time-consuming phase of SDLC because of insufficient communication with stakeholders and the textual nature of requirements. Automation of RE activities, especially requirements quality assessment, is particularly important in the context of agile software development. Very short development cycles of agile projects require quick feedback on the quality of requirements, specified incrementally in the form of so-called "user stories". These are concise, informal descriptions of a system's features or functionalities from the perspective of an end user. They typically follow a structured format, such as "As a [role], I want [feature], so that [benefit]" [24]. Although user stories promote collaboration between stakeholders and development teams, their natural language narrative character makes them still difficult to analyze automatically.

2.2. AI for software requirements engineering

Machine learning (ML) and AI techniques have drawn the attention of the RE community as potential means to automate RE activities and to improve the quality of requirements. Over the past decade, several secondary studies have reviewed AI applications within RE, differing primarily in their breadth and analytical focus. Existing literature can be grouped into two main categories:

1. Comprehensive reviews: providing a comprehensive overview of AI or ML methods across the entire RE process, and
2. Task-specific reviews: addressing specific RE purposes or phases in an abstract form of Bourque and Fairley [25] (characterize; evaluate; explain; understand; predict; improve) and phases of their application (elicitation; analysis; modeling; validation and verification; management; other; NA).

However, the existing secondary studies as shown in Table 1 differ considerably in their focus and scope. Two limitations emerge from the current body of reviews.

First, there is a **conceptual gap**. While several reviews discuss AI support for RE activities in general, none systematically map or compare the quality aspects, datasets, and evaluation strategies employed. Despite extensive surveys on AI applications for RE, there is a lack of systematic investigation into how requirements quality itself is conceptualized and assessed. Existing reviews typically address the automation of RE tasks such as elicitation, classification, traceability, and management, but rarely explore how quality is operationalized within these tasks. For example, while NLP and ML techniques are commonly applied to detect ambiguity, classify requirements, or extract specific elements from user stories [4,8], there is no synthesis of which quality attributes are considered, how they are measured, and which datasets or evaluation strategies are employed. This creates a gap in understanding whether a unifying conceptual framework for AI-driven requirements quality exists or whether studies remain isolated, focusing on individual aspects such as ambiguity, completeness, or template conformance. In short, prior work lacks a structured mapping of quality dimensions, datasets, and evaluation methods across AI-based RE studies, leaving the broader conceptualization of "requirements quality" largely unaddressed.

Second, there is a **temporal gap**. Most systematic reviews and surveys predate the rise of LLMs and GenAI. As a result, the methodological and conceptual shifts introduced by these technologies are not captured. This gap manifests differently across the two main types review categories, into which existing literature reviews can be classified:

- **Comprehensive surveys:** Reviews covering the full RE lifecycle, such as Li et al. [5] and Sonbol et al. [8], analyze trends in AI/ML adoption and the overall automation potential in RE. While these surveys identify the increasing use of DL and embedding-based NLP techniques, they largely end before the widespread integration of LLMs (e.g., BERT, GPT series), which offer richer semantic representations and enable more nuanced quality assessments. Consequently, these studies fail to capture recent methodological innovations such as semantic-level quality evaluation, context-aware reasoning, and generative methods for quality improvement.
- **Phase-specific or task-specific reviews:** Studies focusing on particular RE phases, such as elicitation [1], classification [2,3], or prioritization [26], provide detailed insights into specific AI applications but remain limited in scope. These reviews often report ML and classical NLP techniques applied to narrow quality aspects, without accounting for the broader evolution toward LLMs and GenAI.

2.3. AI for requirements quality assessment

The application of AI/ML techniques for assessing the quality of requirements, and deriving quality improvements has not been addressed by literature reviews published so far. A few surveys on the use of AI/ML for RE in general considered analyzing individual aspects of requirement quality. Zhao et al. [29] and Sonbol et al. [8] provide comprehensive reviews of the applications of NLP techniques for RE. Both studies show requirement analysis being the most and requirement validation (resp. quality assessment) the least NLP-assisted

Table 1
Summary of existing literature reviews on AI in Requirements Engineering.

Author(s)	Year	Timeline of papers	# Papers	Focus	Type
Li et al. [5]	2024	2010–2022	207	Conducted a comprehensive survey on the use of ML in RE, identifying trends and challenges from both academic and grey literature.	Comprehensive
Sonbol et al. [8]	2022	2010–2021	104	Analyzed the application of NLP techniques to change the representation of requirements. Found requirements analysis (incl. classification and traceability) as the most automated task, highlighted the shift from lexical/syntactic features to advanced embeddings including LLMs for semantic-level analysis.	Comprehensive
Raharjana et al. [4]	2021	2009–2020	38	Reviewed NLP applications for specifying user stories in agile development, covering defect discovery, artifact generation, abstraction extraction, and model tracing. Most studies focused on extracting “who,” “what,” and “why” from user stories.	Comprehensive
Mehraj et al. [6]	2024	2017–2023	28	Tertiary study on AI assistance in RE. Identified elicitation, analysis, and classification as most assisted by AI. Specific tasks included ambiguity resolution, completeness and template adherence analysis, and security analysis. Techniques included NLP (PoS-tagging), classical ML (kNN, SVM), DL (CNN), and early LLM applications (BERT).	Comprehensive
Amna and Poels [7]	2022	2001–2021	186	Systematic mapping of user story research. Highlighted gaps: (1) lack of mature solutions for resolving ambiguities early, (2) lack of validation/evaluation. Focused on ambiguity as a quality issue.	Comprehensive
Anwar and Bashir [26]	2023	2000–2021	46	Reviewed AI-based techniques for requirements prioritization (genetic algorithms, fuzzy logic, ant colony optimization). Found ML produces reasonable prioritization results but not always optimal.	Task-specific
Kaur and Kaur [2]	2024	2012–2023	60	Reviewed AI techniques for identifying/classifying functional and non-functional requirements. Transfer learning (BERT) often outperformed traditional ML/DL approaches.	Task-specific
Cheligeer et al. [1]	2022	2007–2021	86	Overview of 15 ML techniques used in requirements elicitation. Focused on automating and improving elicitation from text, comments, app ratings, and evaluating ML/DL performance for elicitation tasks.	Task-specific
Li et al. [27]	2023	2013–2022	26	Analyzed ML-based techniques for requirements traceability. Identified 32 models (mainly traditional ML) for link generation/classification; did not assess requirements quality directly.	Task-specific
Xu et al. [28]	2023	2012–2021	53	Summarized ML methods for requirements management. Focused on demand quality affecting baseline maintenance, traceability, and change management; did not cover DL-based quality assessment.	Task-specific
Liubchenko [3]	2023	2018–2022	24	Discussed ML/DL enhancements of software requirement specifications. Most supported task: classification using NLP, enhanced with RF, SVM, kNN.	Task-specific

phase of RE. Most of the quality assessment approaches are based on the analysis of syntactical and lexical features and focus on such aspects as ambiguity, conformance with templates, or incompleteness. This observation is in line with the result of Kocerka et al. [30], who focus their review on the application of NLP methods for analyzing the quality of textual requirements. Yet, Sonbol et al. observed a shift from syntactical and lexical features toward text embeddings, which allows for semantic-level quality assessment of textual requirements.

2.4. Objective of this review

In summary, existing literature studies the application of AI/ML for RE, but does not take a detailed look at how specific RE tasks are automated using AI/ML. Results of the current literature reviews, which cover publication periods ranging from 2000 to 2023 and include between 24 and 207 analyzed primary studies, indicate that AI/ML applications focus on requirements construction and management tasks. AI/ML-based automation of requirement quality analysis (resp. validation) is limited to detecting predefined specific quality deficits, such as

ambiguous texts. Most commonly employed AI/ML techniques include NLP (also involving DL [31]) and classical ML techniques such as SVM, RF, and kNN. Recent advances in AI/ML, including GenAI, and their applications for RE are captured only by very few of the most recent studies.

As a result, they provide only fragmented insights on requirements quality, addressing isolated aspects such as ambiguity or syntax conformance, and do not capture recent methodological advances that enable richer, semantic-level quality assessments. This study aims to fill this gap by systematically reviewing AI/ML approaches for requirements quality assessment, with two main points of novelty:

1. Temporal novelty: Recent LLM and GenAI developments allow evaluation of quality dimensions that were previously infeasible to assess automatically, such as semantic clarity, consistency, and conceptual completeness. Prior reviews do not reflect these advances.
2. Conceptual novelty: No existing review provides a comprehensive synthesis of quality aspects, associated metrics, and datasets.

Current studies remain isolated, and there is no unifying conceptual basis for comparing or standardizing requirements quality evaluation.

Accordingly, the scope of this SLR is to analyze studies that automatically assess the quality of software requirements by examining quality aspects, datasets, and AI methods. The analysis is agnostic to requirement types, reflecting our focus on how AI automates quality-related RE tasks rather than on the structural form of the requirements themselves. In contrast to prior reviews, this study provides the first systematic mapping of the conceptual foundations of requirements quality within AI-based research, revealing overlaps, divergences, and gaps. In doing so, it addresses the temporal gap left by earlier surveys and establishes a foundation for understanding which quality aspects have been considered, how they are operationalized, which datasets support them, and which AI techniques are applied to their assessment.

3. Research questions

Building on this foundation, the review addresses the following research questions, which guide the SLR:

RQ1: What quality aspects and the associated metrics are considered to assess the quality of software requirements?

We aim at a comprehensive overview of the quality aspects that are recognized in the literature, including their definition and categorization, to explore any overlaps in terminology and meaning across different studies. Additionally, we investigate the way the quality aspects have been derived as well as their exact purposes and the phases of RE they are used. Finally, we examine the metrics defined to measure the quality aspects. We intend to create a catalog of quality aspects and metrics that can be reused, adapted, and extended by RE researchers and practitioners.

RQ2: What datasets were used to create, evaluate, and adjust requirements quality assessment models?

We want to investigate the datasets that have been utilized in the reviewed literature to create, test, and adjust the proposed models for assessing the quality of software requirements. We catalog the datasets, concerning their basic characteristics, such as size, structure, content, availability, type (synthetic vs. real-world), and origin (industrial vs. academic). An additional important aspect we consider is whether, how, and by whom the data has been labeled. This information is crucial for researchers and practitioners to evaluate or replicate the quality of models developed upon different datasets or for creating new models.

RQ3: What AI methods and models are used for the automated assessment of the software requirements' quality?

We aim at exploring what AI methods and models have been utilized in the reviewed studies for the automated assessment of requirements' quality and for other related purposes such as quality improvement. By answering this question, we hope to identify the trends in the usage of AI techniques for validating and improving the quality of software requirements specifications.

4. Review method

Based on the established guidelines for SLRs [32], we conducted the review in distinct phases: identification of relevant data sources and specification of search strategy, identification of inclusion and exclusion criteria, a search for relevant studies, critical appraisal and screening of the studies, data extraction, and synthesis. In the rest of this section, we describe the details of these stages and the methods used.

Table 2
Keywords and synonyms and related Words.

Keyword	Associated words
Requirements Engineering	Requirements Elicitation, requirements analysis, requirements specification, requirements validation, requirements verification, requirements evaluation, requirements assessment, requirements management, requirements traceability, requirements classification, requirements document, requirements fundamentals, requirements process
User Story	Requirement, specification document, requirement document, epic
Deep Learning	Neural networks, deep neural networks, convolutional networks, recurrent networks, deep reinforcement learning, unsupervised learning, supervised learning, machine learning
Natural language processing	natural language, computational linguistics, language models, text processing, text mining, computational language, language understanding, syntactic analysis, semantic analysis
Large language models	large-scale language models, foundation models, generative models, Generative AI, GPT, ChatGPT, LLaMA, BERT, transformer models
Quality	Quality evaluation, quality assessment, quality metric, model evaluation, performance metrics, evaluation criteria, quality indicators, accuracy measures, validation indices

4.1. Data sources

To ensure a comprehensive collection of relevant literature, we included several digital libraries: IEEE Xplore, SpringerLink, ACM Digital Library, the Association for Computational Linguistics (ACL) Library, and the Association for Information Systems (AIS). We executed search queries using the SCOPUS and ScienceDirect databases, which include all the selected libraries. We used SCOPUS due to its extensive coverage; it is the largest database of titles and abstracts, as noted by Kitchenham and Charters in their guidelines for SLRs in software engineering [32]. We additionally used ScienceDirect as a supplementary resource to enhance the thoroughness of our search.

4.2. Search strategy

The search strategy consisted of two phases: automatic search and snowballing. *Automated search* involved executing structured queries on the selected digital library databases. For this purpose, we first derived from the RQs keywords that represented relevant topics (cf. Table 2).

Next, we defined constraints on the review scope, which included publication type, size, time frame, and language. Topics of interest and scope constraints were considered during both the automatic search as part of the search query and during the manual screening of publications (review) as inclusion criteria. Yet, because topic-related inclusion criteria were much more difficult to prove automatically than simple keyword matching, the manual screening focused on these criteria. Fig. 1 presents the final query used for the search.

Snowballing involved forward and backward snowballing [33] on the publications remaining after the full-text review, which was performed by applying the automatic search and screening manually according to the inclusion criteria. Our motivation for the snowballing was to capture relevant publications that could have been potentially missed by automatic search. Whereas the automatic search was performed only once, in the first step, the snowballing was repeated iteratively until no relevant publication could be found. In each iteration, publications identified during snowballing that went through the screening and were classified for the full-text review, were subject to the next snowballing iteration.

```

TITLE-ABS-KEY (
  ("Requirements Engineering" OR "User Story")
  AND
  ({deep learning} OR {NLP} OR {natural language processing}
  OR llm*
  OR "large language model*" OR BERT)
  AND
  (quality))
AND NOT TITLE (review OR "mapping study")
AND (LIMIT-TO (SUBJAREA, "COMP"))
AND (LIMIT-TO (LANGUAGE, "English"))

```

Fig. 1. Search Query for Literature Review. Detailed Boolean search query used to retrieve studies on Requirements Engineering or User Stories in the context of deep learning, NLP, and quality aspects, with filters for subject area (Computer Science) and language.

Table 3

Inclusion criteria (IC).

IC #	Inclusion criteria
IC1	Publications must be from the period between January 2019 and March 2025 to include the most recent research developments.
IC2	Articles must be peer-reviewed to ensure they have undergone evaluation by experts.
IC3	Only primary research articles will be included to capture original research findings.
IC4	Articles must be written in English to ensure consistency in understanding and interpretation.
IC5	Article must be longer than 3 pages, as conference abstracts, workshop summaries, short papers often have.
IC6	Articles must be relevant to the defined search strings and related to the subject areas of our RQs.

4.2.1. Inclusion and exclusion criteria

Studies were eligible for inclusion in the review if they presented topics associated with the RQs in Section 3, that is, studies on the application of AI for assessing the quality of software requirements, including quality aspects and metrics considered, and datasets used. The review included primary research studies, written in English and published in peer-reviewed sources. To ensure that the review includes the most recent developments and research findings in the field, we included articles published between January 2019 and March 2025. We restrict the primary search to publications from January 2019 to March 2025 because this window covers the period when transformer-based LLMs became practically and widely applied in NLP and therefore captures methodological shifts (semantic embeddings, prompt- and model-based evaluations, and generative techniques) not represented in earlier surveys. Table 3 specifies the corresponding inclusion criteria, we used to screen publications during title, abstract, and full-text review.

We did not define exclusion criteria explicitly; implicitly, they can be defined as criteria complementary to the inclusion criteria IC1 to IC6 (c.f. Table 3).

Fig. 2 shows the systematic review process and the number of papers identified at each stage. In step 1 (on March 7, 2025), we conducted an automatic search using the two databases Scopus and ScienceDirect. The search yielded a total of 353 publications, distributed as follows:

- 348 results from Scopus, including 42 preprints.
- 5 results from ScienceDirect, including 3 preprints.

In step 2, we identified and excluded duplicates from the set of initially found publications. We identified a total of 19 duplicate entries, of which 18 occurred within Scopus (17 preprints and one overlapping conference paper) and one from ScienceDirect. After deduplication, 334 unique papers remained.

The inclusion criteria IC1 to IC6 defined in Table 3 were then applied sequentially as step 3:

- **IC1 (Publication period):** 107 publications outside the defined publication period were excluded, resulting in 227 papers.
- **IC2 (Peer review):** 25 non-peer-reviewed papers, all of them preprints, were excluded, reducing the number to 202.
- **IC3 (Primary research):** We excluded 44 non-primary contributions (e.g., conference reviews, errata), resulting in 158 remaining papers.
- **IC4 (Language):** All remaining articles were written in English. No exclusions were necessary.
- **IC5 (Length):** 7 publications with fewer than three pages were excluded, leaving 151 papers.

In steps 4 to 6, we reviewed the remaining publications regarding the relevance of the topics they presented (i.e., according to inclusion criterion IC6 in Table 3). The outcomes were as follows:

- **Step 4. Title screening:** After review of publications' titles, we excluded 29 papers due to the lack of thematic relevance based on title, resulting in 122 papers.
- **Step 5. Abstract screening:** After review of publications' abstracts, we excluded 59 papers, leaving 63 papers.
- **Step 6. Full-text screening:** After review of full publications' text, we excluded 39 papers, resulting in a final set of 24 publications for the next review step of the quality assessment.

4.2.2. Quality assessment

In SLRs, assessing the quality of primary studies is essential to ensuring the validity and reliability of the synthesized findings. Kitchenham and Charter [32] emphasize that quality assessment helps to refine inclusion criteria, interpret study results, and guide future research

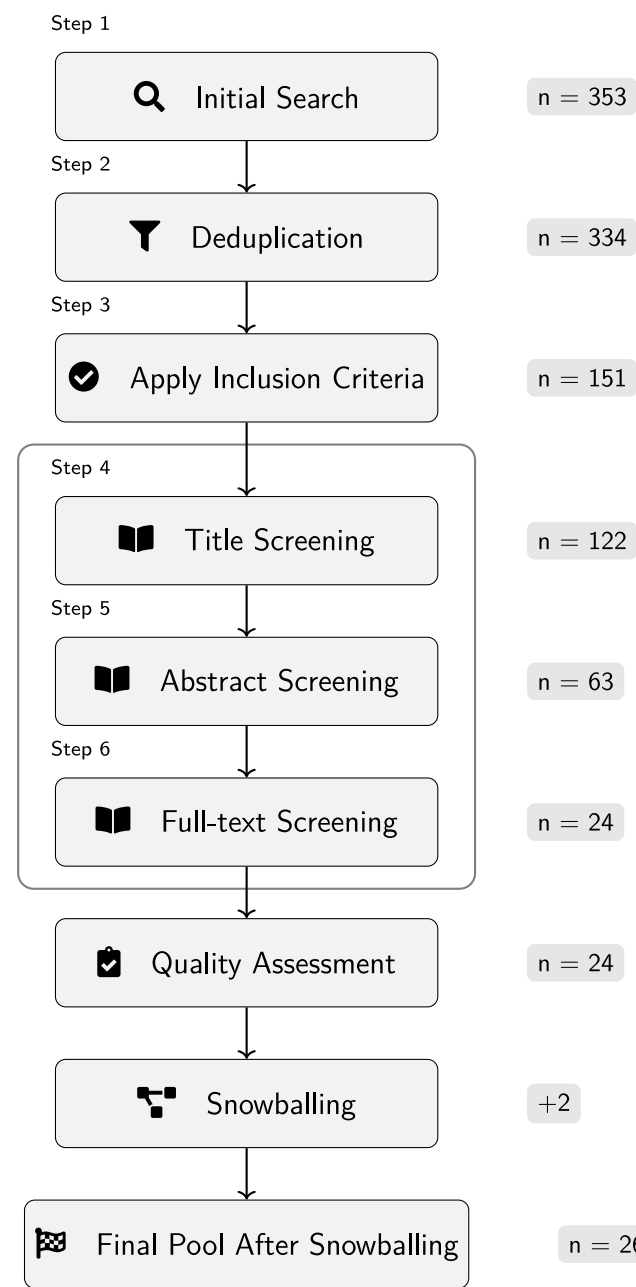


Fig. 2. Search and Selection Process. Flowchart illustrating each step of the study selection, from initial search to final pool after snowballing, including deduplication, screening, quality assessment, and snowballing results.

directions. Although there is no universally agreed definition of study quality, it generally relates to minimizing bias and maximizing both internal and external validity. Internal validity refers to the extent to which a study design prevents systematic errors, while external validity concerns the generalizability of the study outcomes.

We adapted these criteria to evaluate the quality of the publications that passed the screening process (Step 4 – 6 in Fig. 2). Table 4 specifies the detailed quality criteria and the associated rating scales we used in the evaluation.

The publications were compared regarding the sum of scores they obtained on all quality criteria, relative to the highest possible overall score (100%). In case of a publication, for which a specific quality criterion was not applicable and the corresponding rating could not be given (Score = “NA”), the absolute value of the baseline rating of 12

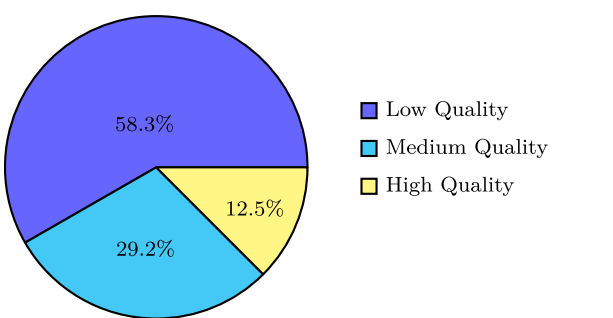


Fig. 3. Quality Assessment of Studies. Pie chart showing the distribution of included studies based on quality ratings: Low, Medium, and High.

points was reduced by the highest score defined for this inapplicable criterion (i.e., two points). This then resulted in a new absolute value for the highest overall score baseline of 100%. We ranked studies, which achieved a total score of at least 85% *high quality* papers, those with a score more than 60% as *medium quality*, and those with at least 40% as *low quality*. We excluded from the study publications that did not achieve the total score of 40% as being of insufficient quality.

After the quality evaluation of the 24 papers included in the full-text review, three of them were classified as high-quality (with two of them achieving the full score of 100%), seven as medium-quality (with scores between 60% and 85%), and the remaining 14 as low quality (with scores down to 40%). None of the publications fell below the 40% inclusion threshold (cf. Fig. 3). We continued the snowballing phase with all 24 initially selected publications.

4.2.3. Snowballing strategy and results

To increase the completeness of our study and include publications not captured through the database search, we applied forward and backward snowballing techniques to the 24 papers selected after full-text screening and quality assessment. All snowball-identified papers underwent the same inclusion process and quality assessment as the original pool.

We performed forward snowballing by querying Scopus and ScienceDirect for newer publications that cited any of the 24 papers in the final pool within the defined publication period. After screening the identified publications and assessing their quality, Paper [34] was included in the study with a perfect quality score.

We performed backward snowballing via Google Scholar by reviewing the publications cited in each of the 24 papers in the final pool. After screening and quality assessment, we included the additional Paper [35], which had a low quality score.

As a result of the snowballing process, two additional relevant papers were identified, bringing the total number of papers included in our review to 26.

4.3. Demographic analysis

To support proper interpretation of the data extracted from the selected publications, we additionally collected their demographic characteristics, specifically the type of the empirical study presented and the context of the study (cf. Table 5).

4.4. Data extraction

To derive the results presented in Section 5, data were extracted from the remaining corpus of 26 included papers in a tabular format. We extracted the following information from each paper:

Table 4
Quality assessment criteria.

ID	Criterion	Scoring
QA1	Research Objectives and Questions Are objectives and RQs clearly specified and appropriate?	(2) Yes: Objective clear; RQs (implicitly) defined (1) Partially: Only objective specified, RQ implicit or partial (0) No: No RQs or objective unclear
QA2	Study Scope and Constraints Is context, scope, limitations on subjects and objects, analysis method, and interpretation of the results explicitly and sufficiently specified?	(2) Yes: Context (industry vs. academia) and scope (type and quantity of subjects and objects) sufficiently described (1) Partially: Both implicit or partial (0) No: Both not described
QA3	Study Design and Methodology Is study design and methodology of data analysis well defined and applied?	(2) Yes: Design and methodology defined, data analysis explicitly described (1) Partially: Design and methodology defined; data analysis unclear (0) No: Only general design and methodology defined
QA4	Replication Package: Raw Data and Results Is data (created or used) described and accessible for replication?	(2) Yes: Data available publicly or in publication (1) Partially: Indirectly accessible or referenced (0) No: Not available (NA): No data considered or confidential (yet should be described)
QA5	Results Presentation Are results systematically analyzed and clearly presented?	(2) Yes: Quantitative analysis process documented and replicable; structured presentation of results in tabular or graphical way (1) Partially: Mixed or incomplete textual description (0) No: No systematic analysis or presentation, not in a systematic way (NA): No results to analyze (e.g., no empirical study)
QA6	Threats to Validity Are threats discussed explicitly and thoroughly?	(2) Yes: All threats addressed in separate section (1) Partially: Discussed, but incomplete or implicit (0) No: Not discussed

Table 5
Demographic characteristics collected for the selected publications.

Study Type
• <i>Literature Review</i>: A systematic summary and analysis of existing literature on a specific topic. This type of study summarizes various findings and perspectives to provide a comprehensive understanding of the subject matter. • <i>Experiment</i>: An empirical investigation to observe the effects of different variables to establish relationships between them. • <i>Case Study</i>: An examination of a small number of cases to gain insights into a specific issue. • <i>Survey</i>: A method of data collection where a sample of individuals is asked questions to gather information about their opinions, behaviors, or characteristics. • <i>Theoretical Paper</i>: A study that develops or discusses theoretical models or concepts without primarily relying on empirical data.
Study Context
• <i>Industrial</i>: Research conducted in an industrial or commercial setting, often focusing on practical applications, process improvements, or product development. • <i>Academic</i>: Research carried out in an academic environment, such as universities or research institutes, with a primary focus on advancing knowledge, developing new theories, or contributing to scientific discourse.

- Extraction of named quality aspects and their definitions as stated in the papers. This includes identifying the quality aspects addressed by each paper and defining them as provided within the text.
- Extraction of the development of these quality aspects and identification of the basis of their development, including the names of standards or guidelines referenced in the papers.
- Classification of assessment method regarding (1) the purpose, based on the purpose-taxonomies of empirical studies [25] and of data analysis [36] (characterize; evaluate; explain; understand; predict; improve), and (2) the RE phases, in which assessment is performed (elicitation; analysis; modeling; validation and verification; management; other; NA).
- Extraction of the metrics used for evaluating the quality aspects. This involves identifying the specific metrics employed by each paper and their relevance to the evaluation of quality aspects.
- Extraction of the datasets used in the papers, including information on their size, structure, development, and labeling process, if applicable. This provides insights into the data used for experimentation and analysis in the papers.
- Extraction of NLP/DL/ML/AI methods applied in the papers to fulfill their objectives. This includes identifying the specific methods employed and their application in addressing the research objectives of the papers.
- Classification of the application of these methods based on their effectiveness on the data. This involves evaluating the effectiveness of the methods in achieving their intended purposes and their impact on the quality of the data analyzed.

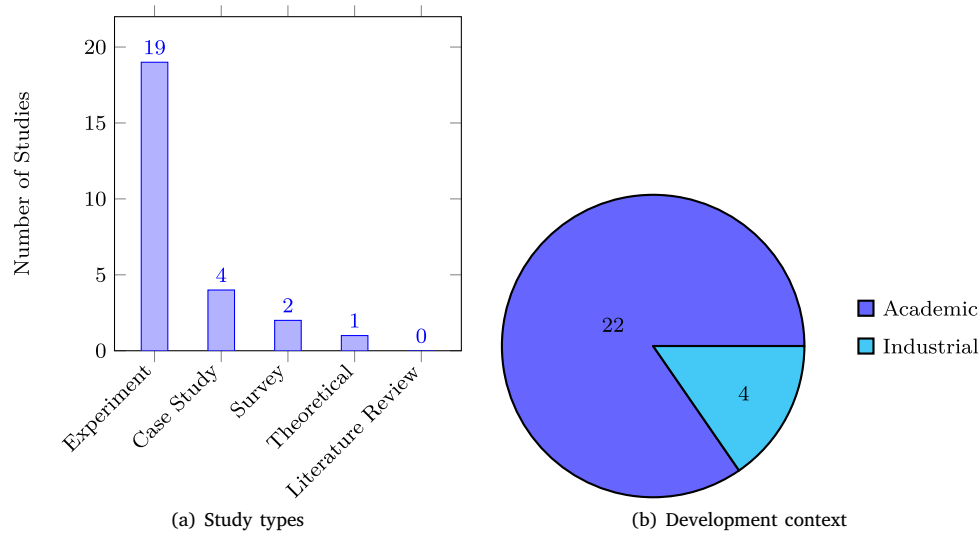


Fig. 4. Descriptive Study Demographics. (a) Bar chart of study types including Experimental, Case Study, Survey, Theoretical, and Literature Review; (b) Pie chart depicting the development context of studies as Academic or Industrial.

5. Results

This section presents the results of the review of the 26 publications. First, we summarize the demographic characteristics of the published studies and then the findings we extracted regarding our three RQs.

5.1. Studies overview

Fig. 4 shows the demographic characteristics of the 26 publications included in the review. Of these, 19 reported experiments, four were case studies, two were surveys, and one was a theoretical paper; none were literature reviews. In terms of development context, most studies (22 out of 26) were conducted in academic settings, while four originated from industrial or applied environments (cf. Fig. 4).

5.2. RQ1: What quality aspects and associated metrics are considered to assess the quality of software requirements?

To comprehensively address RQ1, we structured our analysis into three subdimensions. First, we identify and categorize the quality aspects addressed across the selected studies and explore their conceptual overlaps in Section 5.2.1. Second, in Section 5.2.2 we map the identified aspects to their practical purposes and the phases of RE in which they are applied. Third, we examine the evaluation metrics used to assess these quality aspects and reflect on their methodological implications in Section 5.2.3.

5.2.1. Quality aspects: Origins and usage frequencies

Table 6 provides a structured overview of the quality aspects and their definitions, sorted by the number of papers in which they were considered.

A broad range of quality aspects under varying names is discussed in the literature, including both *semantic* and *syntactic* quality. *Semantic quality* refers to the meaning and intent of the requirements. It assesses whether the requirements accurately capture what the user needs and can be correctly interpreted by stakeholders. Example aspects of semantic quality include *Completeness*, *Correctness*, *Testability*, or *Ambiguity*. *Syntactic quality*, on the other hand, refers to the structure and form of the requirements. It evaluates whether the requirements follow the correct grammar, format, and notation rules. Example aspects of semantic quality include *Complexity* and *Conciseness*.

The definitions of each quality aspect vary significantly in granularity and emphasis. Some papers offer formalized or operational definitions (e.g., *Testability* is defined by its ability to support test case derivation in Li et al. [37]); others rely on implicit or example-based characterizations. In some cases, the same term is used with different underlying meanings. For instance, *Clarity* is understood in Rahman et al. [34] as a combination of acceptance criteria regarding textual understanding as well as giving value to their target audience, split into technical versus non-technical stakeholders.

Several quality aspects were used inconsistently across studies, sometimes under different names but with similar definitions, or vice versa. For example, *Ambiguity*, *Vagueness*, and *Clarity* frequently overlap conceptually, all referring to potential misinterpretations or lack of precision. Similarly, *Atomicity* and *Independency* all contribute to the notion of manageable and self-contained user stories. Conversely, the same label can be defined differently across papers, e.g., *Estimability* is described in Paper [46] as technical feasibility, while Paper [37] emphasizes team understanding and effort estimation, aligning more with predictability and cost approximation.

In terms of origin, the quality aspects come from various sources. Many aspects are grounded in recognized standards and guidelines. For instance, ISO standard ISO/IEC/IEEE 29148:2018 [41] together with Wilson et al. [40] form a comprehensive set of relevant quality aspects. Among the aspects we identified in our review, only *Independence*, *Usability*, and *Estimability* are not defined in these two sources. On the other hand, well-established specific frameworks such as INVEST by Wake [45] and QUS by Lucassen et al. [20] served as sources of quality aspects such as *Independency*, *Negotiability*, *Usability*, *Estimability*, *Conciseness*, and *Testability*.

In our analysis, we distinguish between two types of sources that define quality aspects: *empirical* and *conceptual*. We define *empirical* quality aspects as those derived from observed issues in real-world RE practice or through empirical studies. These aspects are typically rooted in problems identified during the formulation, interpretation, or validation of user stories. For example, *Ambiguity* has been widely reported as a practical problem, leading to concrete subcategories such as *Ambiguous Adverbs and Adjectives* and *Anaphoric Ambiguity*, as found in Paper [47] and Paper [48].

In contrast, *conceptual* quality aspects are derived through theoretical reasoning or formal modeling of syntactic or semantic characteristics of user stories. These often reflect linguistic or structural

Table 6

Quality aspects of user stories in literature.

Aspect	Names in the studies (Count)	Count	Definition by studies (if available)	Reference for definition	Type
Ambiguity	Unambiguity (10), Ambiguity (2), Ambiguous Adverbs and Adjectives (2), Anaphoric Ambiguity (2), Ambiguous Element (1), Loopholes (1), Number of Conjunctions (1)	19	Unambiguous: avoids terms leading to multiple interpretations; fuzzy requirements require domain-specific vocabulary; only one interpretation	[38], [39], [40], [41], [20], [42], [43], [44]	Semantic
Clarity	Understandable (3), Subjective Language (2), Superlatives (2), Comparatives (2), Incorrect Goal Semantics (1), Incorrect Softgoal Semantics (1), Readability (1), Specificity (1), Readability Statistics (1)	14	User stories should be clear, measurable; should be easy to read for all stakeholders	[40], [41], [43]	Semantic
Structure	Well-formed (1), Full Sentence (1), Punctuation Marks (1), Misspellings (1), Conforming (1), passive voice (1), negative statements (1), Imperative (1), Directives (1), Incorrect Actor Syntax (1), Incorrect Goal Syntax (1), Incorrect Softgoal Syntax (1), Incorrect Task Syntax (1), Incorrect Resource Syntax (1)	14	A user stories includes at least role and means; is a well-formed full sentence; follows a standard template	[40], [20], [41], [43]	Syntactic
Testability	Testable (4), Verifiable (4), Validatable (1)	9	Must provide information to make test development possible; fulfillment can be proven or measured	[40], [45], [41]	Semantic
Completeness	Complete (4), Completeness (2), Incompleteness (1), Incomplete Text (1), Implicit Words (1)	9	All information needed to understand the requirement is included; incomplete: incompletely expressed or with unplanned content	[41], [38], [40], [44]	Semantic
Conciseness	Small (3), Conciseness (1), Minimal (1), Size (1), Element Size (1)	7	Minimal: User stories contain only a role, means, and ends; user stories are of the right size	[41], [40], [20], [45]	Syntactic
Consistency	Consistency (3), Internal Consistency (2), Inconsistency (1), Contradiction (1)	7	Negative contradiction, scope limitation	[41], [44]	Semantic
Vagueness	Vague Pronouns (2), Vagueness (1), Weak Phrases (1), Vague Words (1), Open-ended Non-Verifiable Term (1)	6		[40], [43]	Semantic
Independence	Independent (3), Conflicting Goals/Tasks (1), Goal/Task and Subgoal/Subtask Mismatch (1)	5	Can be delivered without depending on others	[45]	Semantic
Negotiability	Negotiable (3), Modifiable (1), Options (1)	5	Leaves space for discussion; not a strict contract	[40], [45]	Semantic
Usability	Valuable (3), Usable (1), Technical Aspect (1)	5		[45]	Semantic
Estimability	Estimatable (4)	4	User stories is technically achievable; should be estimable, understandable enough to size	[20], [45]	Semantic
Correctness	Correct (3), Accurate (1)	4	The need is accurately represented in the requirement	[41], [40]	Semantic
Atomicity	Singular (1), Atomic (1), Conceptually Sound (1), Too Specific Element (1)	4	Atomic: exactly one feature; defines only one aspect	[41], [20]	Semantic
Feasibility	Feasible (1), Security Statistics (1), Ranked (1)	3		[40], [41]	Semantic
Redundancy	Non-Redundancy (2), Similar Elements (1)	3		[41]	Syntactic
Traceability	Traceable (1), Continuances (1)	2		[40]	Semantic
Complexity	Complex Sentences (1), Complexity (1)	2		[40]	Syntactic
Appropriateness	Appropriate (1), Necessary (1)	2	Requirement excludes unnecessary constraints, is realizable	[41]	Semantic

correctness as proposed by the respective authors. For instance, Paper [49] identifies several syntactic issues such as *Incorrect Actor Syntax*, *Incorrect Goal Syntax*, *Incorrect Software Syntax*, and *Incorrect Task Syntax*, all of which are based on a structural conceptualization of what constitutes a syntactically well-formed user story. Similarly, *Superlatives* and *Comparatives*, as discussed in Paper [47], are seen as conceptually problematic due to their imprecise nature in requirements.

A particular case can be made for aspects such as *Structure* and *Redundancy*, which are frequently mentioned in various names in the literature but rarely defined in a standardized way (cf. Table 6). We

define *Structure* as the syntactic organization of a user story, including adherence to expected patterns such as the “As a [role] I want [goal] so that [reason]” format. This category includes issues such as *Incorrect Actor Syntax*, *Passive Voice*, *Misspelling*, and *Punctuation Marks*, identified in Papers [47,50], and [51]. *Redundancy* is defined as the unnecessary repetition or duplication of information across or within user stories, potentially harming the clarity and conciseness of those. Relevant examples include *Non-Redundancy* from Paper [52] and *Similar Elements* from Paper [49].

This structured differentiation between empirical and conceptual origins highlights the heterogeneous nature of quality criteria in user

story formulation. It also illustrates how syntactic quality is informed by both practical experience and theoretical reasoning.

5.2.2. Quality aspects: Usage purposes and RE phases

To explore the role of specific quality aspects and the RE lifecycle stages in which they are assessed, we also investigated the purposes for which assessments were performed. Our analysis was motivated by the observation that, depending on the assessment purpose, different aspects of quality should be considered, for instance, detailed sub-aspects or factors that may influence quality. We distinguished four main purposes of quality assessment based on the taxonomies of empirical studies [25] and of data analysis. [36]:

- *Characterize/Evaluate* purpose corresponds to the descriptive analysis and aims at answering the question *What is/was the quality of requirements?*
- *Explain/Understand* purpose corresponds to diagnostic analysis and aims at answering the question *Why is the quality of requirements as observed?*
- *Predict* purpose corresponds to predictive analysis and aims at answering the question *What is the quality of requirements likely to be?*
- *Improve* purpose corresponds to prescriptive analysis and aims at answering the question *What can/should be done about the quality of requirements to improve it?*

Table 7 shows a mapping of each quality aspect to one or more purposes and their corresponding integration into typical RE phases [25]: *Elicitation, Analysis, Modeling, Verification and Validation, and Management*.

Quality *Characterization/Evaluation* is a dominating purpose of defining specific quality aspects and encompasses all aspects we identified in the study. In total, 19 out of 19 aspects were mentioned at least in one publication in the context of requirement quality characterization/evaluation, making it the most prevalent intent. The next most common purposes are *Explanation/Understanding* and *Improvement*, for which 12 quality aspects were used, respectively. To explain/understand requirements quality, particularly conceptually, abstract or context-dependent aspects were used, such as *Testability, Conciseness, and Clarity*. In the context of improving the quality of requirements, quality aspects such as *Usability, Negotiability, and Testability* were used. The *Prediction* was rarely named as the usage purpose and referred to only four aspects.

Regarding the phase of the RE lifecycle in which quality aspects are used, the *Analysis* phase is clearly dominant. It is associated with 17 of the 19 quality aspects, e.g., *Ambiguity, Testability and Consistency*. The *Elicitation* phase follows in relevance, with 15 aspects considered, particularly those concerned with early-stage expression and clarity, such as *Testability, Conciseness, and Negotiability*.

Later RE lifecycle phases are considerably less represented. The *Modeling* phase appears only in connection with the aspects *Correctness, Usability, and Ambiguity*, indicating that most quality assessment frameworks are not yet integrated into formal modeling activities. Similarly, the *Validation and Verification* phase is linked to only 10 aspects, including *Structure, Atomicity, or Redundancy*, and is often considered as a secondary phase after the analysis. The *Management* phase is not mentioned at all, indicating a potential gap in managing requirements quality across the entire RE lifecycle.

Overall, the mapping illustrates the clear trend that most quality aspects are used to support early phases of RE, specifically *Analysis* and *Elicitation* tasks, with fewer targeting downstream phases or aiming at proactive improvement or prediction of requirements quality. This suggests a strong orientation of current research on “building-in” quality and shows the potential of extending it to quality analysis (i.e., “testing-in” quality) in the *Validation* and *Management* phases of RE.

5.2.3. Quality metrics and quality measurement

To understand how the identified quality aspects were operationalized in empirical studies, we analyzed the associated quality evaluation metrics used across the included papers and summarized them in Table 8.

The most prominent evaluation method, considered in 14 papers, was the use of *binary metrics* (c.f. Table 8). This generic measurement method involves assessing on a “Yes/No” scale the presence or absence of specific quality violations or compliance with defined criteria, such as the fulfillment of INVEST principles, the presence of weak phrases, or the detection of ambiguity.

Several papers (e.g., Papers [34,37,46]) employed similarly generic metrics based on *5-point Likert scales*, which measured the level of requirement’s compliance to specific quality aspect such as *Consistency* and *Conciseness*, or to a set of quality aspects such as INVEST. These subjective rating scales were often used in the context of expert evaluations or user satisfaction studies.

Less frequently, we found specialized and task-specific metrics. For example, Paper [61] utilized *translation accuracy, confidence scores, and translation loops* to evaluate the machine translation of user stories into formal representations. Similarly, Papers [48,59] introduced a nuanced metric of *full vs. partial matching* to assess the resolution of *anaphoric ambiguity*.

Other rare but notable metrics included *text structure, semantic similarity, and defect count* used for contradiction detection. Paper [66,67] reported using no explicit metric at all, relying instead on narrative or qualitative analysis.

Across the examined corpus, overlaps were observed primarily among those quality aspects that had a textual or linguistic focus. For example, *Ambiguity* was frequently evaluated using *binary metrics*, while *Testability* and *Conciseness* have *Likert-based* subjective ratings. This suggests a convergence in how such aspects are perceived and operationalized in the literature, despite their conceptual differences.

5.3. RQ2: What datasets were used to create, evaluate, and adjust requirements quality assessment models?

Table 9 presents an overview of the datasets considered in the reviewed papers to train, validate, or evaluate models related to quality aspects of requirements artifacts. We identified a total of 35 datasets distributed across the 26 publications.

However, not all papers made use of a dataset. Paper [67] did not rely on any dataset at all. Conversely, several papers employed multiple datasets. Specifically, the seven papers [47,49,51,54,59,64,65] each used two distinct datasets. The two papers [34,48] utilized three datasets each, while Paper [68] made use of four different datasets. The remaining papers used a single dataset.

In the following paragraph, we briefly summarize the distribution of the datasets regarding their several basic properties.

Availability. Among the datasets, 17 were proprietary and inaccessible to the broader research community, while 18 were publicly available. This relatively balanced distribution underscores a mix of academic transparency and industry confidentiality within the field.

Synthetic vs. Real-World Data. Out of the 35 datasets we identified, 20 were synthetic, and 15 were derived from real-world projects. Synthetic datasets were often constructed in academic contexts for specific quality assessment scenarios, while real-world datasets were typically obtained from industry projects, involving companies like Daimler in Paper [53], or domains such as healthcare in Paper [49] and finance in Paper [34].

Labeling. The majority of the datasets (26 out of 35) were labeled to facilitate supervised learning or evaluation. Of these, 25 were manually annotated by domain experts, while one was labeled using automatic or self-supervised methods. Ten datasets remained unlabeled.

Table 7
Mapping of quality aspects to purpose and phase (with frequency counts).

	Purpose				Phase				
	Characterize, Evaluate	Explain, Understand	Predict	Improve	Elicitation	Analysis	Modeling	Verification-Validation	Management
Ambiguity	15	1	1	5	2	10	1	4	
Appropriateness	1								
Atomicity	3					2		1	
Clarity	5	1		1	1	3		1	
Completeness	7	1		2	2	4		2	
Complexity	2				1			1	
Conciseness	5	2		3	4	4		1	
Consistency	5	1		2		5		1	
Correctness	4	1		2	1	2	1		
Estimability	2	1		2	3	3			
Feasibility	2				1	1			
Independence	2	1	1	2	3	2		2	
Negotiability	2	1		2	4	2			
Redundancy	3	1		1		2		1	
Structure	5		1		1	3		1	
Testability	5	3		3	4	5			
Traceability	1				1				
Usability	3	1		3	3	3	1		
Vagueness	3		1		1	3			

Table 8
Overview of evaluation metrics and their frequency.

Metric	Number of papers	Papers
Binary (Yes/No)	14	[35,47–51,53–60]
5-Point Likert Scale (General)	2	[37,52]
Average Likert Rating (Satisfaction/Quality)	2	[34,46]
Translation Accuracy	1	[61]
Confidence Scores	1	[61]
Translation Loops	1	[61]
Semantic Similarity	1	[62]
Defect Count	1	[63]
Text Structure Analysis	1	[64]
Specification Depth	1	[64]
Readability Index	1	[64]
Number of Content Clusters	1	[65]
None (Qualitative Only)	2	[66,67]

Development Context. 15 datasets were developed in industrial contexts, while 20 originated from academic environments. Industrial datasets often reflect real-world systems and tend to be more extensive, whereas academic datasets are typically designed for controlled experimentation and ease of access.

Structure and Content. Regarding the type of data content, 18 datasets focused on requirements in general, 5 contained user stories, two provided full sentences of specification documents, two provided the specification documents as a whole, and one provided the pages of a specification document separately. The number of entries ranged widely:

- **Requirements:** 10 to 15,000 entries (e.g., *Electric Bus* [54], *DAMIR* [48,59]).
- **User stories:** 6 to 9060 entries (e.g., *QUS framework* [50]).
- **Specification documents:** 1 to 3 entries (e.g., *Mechanical Lung Ventilator* [60]).
- **Sentences:** 6757 to 34,268 (e.g. *PURE* dataset [47,64,65])
- **Pages:** 7 (e.g. *Club Management Portal* [52])

Frequently Used Datasets. The dataset *PURE* appeared most frequently, being used in four papers, where Paper [47] of them only used one project of the whole *PURE* dataset. Another recurring dataset

is *PROMISE*, referenced in Paper [55,57]. The datasets *DAMIR* and *ReqEval* were used by Paper [48,59]. These datasets served as reference points for evaluating the generalizability of quality models.

Comparability. While a few datasets like *PURE* or *PROMISE* were reused across multiple papers, the majority of datasets were used solely in one specific study. Differences in structure, labeling schema, and quality aspects annotated limit direct comparability between the results of the studies, particularly the performance of the requirements quality assessment methods and models. Nonetheless, publicly available and expert-labeled datasets like *DAMIR* and *QUS* offer potential for benchmarking future approaches.

5.4. RQ3: What AI methods and models are used for the automated assessment of the software requirements' quality?

To address RQ3, we structured the analysis into three subdimensions. First, we provide an overview of the AI methods applied across the included studies, with a focus on their data preprocessing and model creation components in Section 5.4.1. Second, in Section 5.4.2 we investigate how the quality of the created requirements quality assessment models was evaluated, both in terms of evaluation metrics used and the methodological robustness of evaluation strategies. Third, we identify trends in the use of AI methods over time, highlighting

Table 9

Overview of named datasets used in RE papers.

Dataset name	Type	Availability	Context	Dataset size/structure	Labeled
PURE [47] [64] [65]	Synthetic	Public	Industry	34,268 sentences	Yes
DigitalHome (PURE) [51]	Real	Proprietary	Academic	63 requirements	Yes
Stopwatch [51]	Real	Proprietary	Academic	10 requirements	Yes
Wikipedia (Computer Science Text corpora) [65]	Synthetic	Public	Academic	Not specified	No
Izhar et al. (2025) (PROMISE) [57]	Synthetic	Proprietary	Industry	1553 requirements (976 from PROMISE)	Yes
PROMISE [55]	Synthetic	Public	Industry	9060 user stories	Yes
University alumni [49]	Synthetic	Public	Academic	Not specified	No
Healthcare system [49]	Synthetic	Public	Academic	Not specified	No
Daimler [53]	Real	Proprietary	Industry	117 requirements	No
Berhanu and Alemneh (2023) [35]	Synthetic	Public	Industry	3100 requirements	Yes
Meteo-satellite [66]	Real	Proprietary	Industry	13,000 requirements	No
Cosler et al. (2023) [61]	Synthetic	Proprietary	Academic	36 requirements	Yes
Coffee Machine [68]	Synthetic	Public	Academic	21 requirements	Yes
E-shop [68]	Synthetic	Proprietary	Academic	33 requirements	Yes
Library [68]	Real	Proprietary	Industry	109 requirements	Yes
ETCS [68]	Real	Proprietary	Industry	64 requirements	Yes
Gärtner and Göhlich (2023) [58]	Synthetic	Proprietary	Industry	1071 requirements	No
Electric bus [54]	Real	Proprietary	Industry	1561 requirements	Yes
Fischbach et al. (2020) [54]	Synthetic	Public	Academic	15,000 requirements (300 used)	Yes
Habib et al. (2021) [47]	Synthetic	Proprietary	Academic	3246 requirements	Yes
ReqEval [48] [59]	Synthetic	Public	Academic	98 requirements	Yes
DAMIR [48] [59]	Synthetic	Public	Academic	737 requirements	Yes
CoNLL2011 [48]	Synthetic	Public	Academic	6757 sentences	Yes
ARM (reconstructed) [64]	Real	Public	Academic	Not specified	No
Club management portal [52]	Synthetic	Public	Academic	7 pages	Yes
Dalpiaz et al. (2019) [37]	Real	Public	Industry	186 user stories	Yes
Mechanical Lung Ventilator [60]	Real	Proprietary	Industry	1 Specification Document	Yes
IOS mobile application [34]	Real	Proprietary	Academic	Not specified	Yes
Insulin Pump, Mental Health Care System, Weather Station (Sommerville) [34]	Synthetic	Public	Academic	Not specified	Yes
Finance Management System, Wellness App [34]	Real	Public	Industry	Not specified	Yes
QUS framework [50]	Real	Public	Academic	3468 user stories	Yes
SDMP [62]	Synthetic	Proprietary	Academic	Not specified	No
Seifert et al. (2025) [63]	Real	Public	Academic	3 specification documents	Yes
Zhang et al. (2023) [56]	Real	Proprietary	Industry	215 requirements	No
Zhang et al. (2024) [46]	Synthetic	Proprietary	Industry	6 user stories	Yes

Table 10

Frequency of AI methods used across the included studies.

Paper	Preprocessing			Modeling													
	NLP	BERT	GPT	GPT	Llama	Other LLMs	LR	MLP	RF	DT	NB	kNN	SVM	CNN	GenAI	NLI	k-means
LLM-Centric Direct Modeling																	
[68], [50], [62], [58], [60]				✓													
[61]						✓											
[52]				✓		✓											
[46]						✓											
[37]				✓	✓	✓											
[63]				✓		✓											
[51]					✓												
[67]				✓											✓		
[49]	✓															✓	
LLM-Centric Pipelines																	
[55]		✓		✓										✓			
[59]		✓		✓	✓	✓											
[34]		✓		✓													
Classical-ML Pipelines																	
[35]	✓						✓			✓	✓	✓	✓				
[54]	✓										✓	✓					
[47]	✓							✓				✓	✓				
[57]	✓								✓	✓	✓	✓	✓				
[64]	✓						✓			✓	✓		✓				✓
Mixed Pipelines																	
[53]		✓	✓									✓					
[48]		✓	✓										✓				
[65]		✓															✓
[66]	✓		✓												✓		
[56]		✓													✓		

shifts in model types, application purposes, and evaluation practices in Section 5.4.3.

5.4.1. Overview of AI-based quality assessment methods

Across the 26 included papers, various AI methods, specifically DL and NLP methods, were employed to assess the quality of software artifacts such as requirements, user stories, or specifications. Table 10 illustrates the frequency of use of major AI methods.

The most frequently used categories of methods are:

- **LLM-Centric Direct Modeling (GPT-/LLM-focused):** This group contains 13 papers [37,46,49–52,58,60–63,67,68], characterized by the use of LLMs such as *GPT-3*, *GPT-3.5*, *GPT-4 Turbo*, *CodeL-LaMA34b*, *Bloom*, *Nous Hermes 2*, *Phi-3-medium*, and *LLaMA* variants. Notably, Paper [67] suggests integrating *GPT-4 Turbo* within its pipeline despite originally presenting a methodological framework. Paper [49] employs an Xtext grammar-based approach (TGRL) for generating graphical GRL syntax, combining it with *Natural Language Inference (NLI)*, positioning it within this LLM-centric pipeline due to its primarily NLP-driven methodology. These studies leverage LLMs for tasks including requirements rewriting, critique generation, evaluation, and multi-agent LLM architectures by Paper [46].
- **LLM-Centric Pipeline (BERT-focused Preprocessing):** Three papers [34,55,59] leverage transformer-based models such as *BERT* predominantly for preprocessing, converting textual requirements into rich contextual embeddings that serve as inputs for subsequent modeling. For example, [55] uses *BERT* embeddings combined with *GPT*-based generation to enhance requirements classification, while [59] applies a combination of *BERT* and other LLM embeddings (including *LLaMA* and *Mistral*) for more nuanced semantic representations. The paper [34] focuses on the use of *BERT* embeddings to improve semantic understanding of requirements, facilitating more accurate quality assessments.

- **Classical-ML Pipeline:** Five papers [35,47,54,57,64] primarily employ traditional ML algorithms such as *SVM*, *Logistic Regression (LR)*, *RF*, *Decision Trees (DT)*, *kNN*, and *Naïve Bayes (NB)*. For instance, Paper [47] contrasts *SVM* and *MLP* approaches and assigns itself to this category due to the superior performance of *SVM*. These models utilize classical NLP preprocessing techniques (e.g., *tokenization*, *TF-IDF*, *bag-of-words*) and serve to classify or score quality attributes in requirements.
- **LLM-Centric Preprocessing + Classical ML Pipeline:** A smaller subset of papers [48,53,65] adopt a mixed strategy by using LLM-based preprocessing (notably *BERT* embeddings and *GPT*-based vectorization) combined with classical ML classifiers. For example, [53] employs cosine similarity-based *kNN* within its *KATE* algorithm to enhance context-aware example selection. Paper [48] utilizes *BERT*, *DistilBERT* and *DeBERTaV3* to formulate embeddings.
- **Deep Learning (Non-Classical ML) Pipeline:** Two papers [56, 66] focus on deep neural networks like *CNNs* and *Multilayer Perceptrons (MLPs)* without leveraging transformer-based LLMs as core models. These approaches emphasize feature extraction through *CNN* architectures and non-transformer DL for quality prediction.

Table 11 provides a consolidated view of the mapping between AI methods and the specific quality aspects they are applied to. These aspects serve as a structured lens to organize the analysis of AI applications in requirements quality assessment. It should be noted that the derivation and justification of these quality aspects, based on a systematic analysis of the literature, are presented in detail in Section 5.2.1 (RQ1). Table 11 complements the detailed frequency overview in Table 10 by highlighting the preprocessing and modeling techniques that are typically employed for each quality dimension.

Table 11
Mapping of quality aspects to commonly used AI methods.

Quality aspect	Preprocessing methods	Modeling methods
Ambiguity	NLP, BERT, RoBERTa, SBERT, GPT	CNN, SVM, kNN, LLaMA, GPT, Decision Trees, Naïve Bayes, NLI, Codex, Bloom, google-flan
Completeness	NLP, BERT	CNN, GPT, LLaMA, GenAI, NLI
Correctness	NLP	GPT, LLaMA, Codex, Bloom
Testability	NLP, BERT	CNN, GPT, LLaMA, multiagent system
Clarity	NLP, BERT	MLP, SVM, kNN, Naïve Bayes, Logistic Regression, NLI
Conciseness	NLP	GPT, LLaMA, NLI, multiagent system
Atomicity	–	GPT, LLaMA, NLI
Consistency	NLP, RoBERTa	GPT, CNN, kNN, Naïve Bayes
Traceability	NLP	–
Appropriateness	–	LLaMA
Structure	NLP	LLaMA, kNN, SVM, Decision Trees, Naïve Bayes, Logistic Regression, NLI
Estimability	–	GPT, LLaMA, multiagent system
Independence	–	GPT, LLaMA, multiagent system, NLI
Negotiability	NLP	GPT, LLaMA, multiagent system
Vagueness	NLP	MLP, SVM, kNN, Decision Trees, Naïve Bayes, Logistic Regression
Complexity	NLP	NLI
Usable	BERT	GPT, LLaMA, multiagent system
Feasible	NLP	LLaMA
Redundancy	RoBERTa	GPT, CNN, NLI

LLM-Centric Direct Modeling (GPT-/LLM-focused) includes methods that directly apply LLM methods to assess the quality of requirements without their prior preparation. These approaches predominantly address semantically rich and open-text quality dimensions such as *Ambiguity* in Paper [49,51,52,61], *Conciseness* in Paper [37,46,52], and *Testability* in Paper [37,46]. Leveraging strong contextual understanding and generative capabilities, these models identify vague or redundant formulations, suggest clearer rephrasings, and assess whether requirements are verifiable. For instance, studies such as [52,61], and [37] apply LLMs with minimal preprocessing directly to textual requirements for quality assessment and enhancement. Less commonly addressed quality aspects such as *Estimability* and *Negotiability* are typically handled within broader, often multi-agent or LLM-driven frameworks (e.g., [37,46]), highlighting emerging directions for AI-supported RE.

LLM-Centric pipelines (BERT-focused) consist of preprocessing and modeling phases, both using LLM models. The preprocessing uses transformer models to combine deep contextual embeddings as an intermediate representation to capture nuanced semantic features before the modeling phase. Papers [34,55,59] utilize embeddings from BERT or RoBERTa to represent requirements text as dense vectors, which are then analyzed or classified for quality issues such as *Ambiguity* in Paper [59], *Clarity* in Paper [34], and *Consistency* in Paper [55]. This hybrid approach balances the rich semantic encoding of transformers with the generative or discriminative power of LLMs or classical ML classifiers.

Classical-ML pipelines employ NLP techniques to extract relevant features from requirements specifications, which are then analyzed using classical ML techniques such as SVM, kNN, or RF. Extracted features typically include structural properties of requirements that indicate higher-level quality aspects, such as, for example, *Vagueness* [35,47], *Consistency* [54], *Ambiguity* [57], and *Structure* [64]. These methods rely on handcrafted or statistical features such as bag-of-words, TF-IDF, or unigram embeddings to convert textual data into numeric vectors for classification or regression. For example, [47] applies TF-IDF weighted

embeddings combined with SVM for *Vagueness* detection, while [57] uses RF on TF-IDF features to identify complex *Ambiguity* patterns.

Mixed pipelines include approaches that combine classical-ML and LLM techniques within a preprocessing-modeling pipeline. Preprocessing involves using LLMs or transformer embeddings to convert textual requirements into rich vector representations, which then serve as input features for classical ML classifiers such as SVM or RF. Papers [48,53,65] demonstrate this pipeline, targeting all the quality concerns of *Ambiguity*, where transformer-based embeddings enable improved feature extraction before classification. Modeling with DL techniques (e.g., CNNs, RNNs) showed promising results in capturing local and sequential textual patterns relevant to the quality dimension *Completeness* [56,66].

In summary, LLM-centric direct modeling excels in semantic-rich and open-ended quality aspects like *Ambiguity*, *Conciseness*, and *Testability*. Transformer-embedding-based pipelines provide a middle ground, enabling richer semantic feature extraction. Pipelines based on classical-ML techniques are effective for structural and surface-level qualities, emphasizing explainability and efficiency. Pipelines involving DL models offer advantages in pattern recognition for *Consistency* and *Completeness*.

5.4.2. Evaluation of the AI-based quality assessment methods

Of the 26 reviewed papers, 24 provided some form of empirical evaluation of the proposed AI-based approaches to assess the quality of the requirements. Paper [61,67] did not use any evaluation metrics. To provide a clear overview of the evaluation approaches applied across the reviewed studies, Table 12 summarizes the different evaluation metrics along with the corresponding studies that employed them.

The applied evaluation metrics can be grouped into five categories, reflecting both the type of AI method used and the nature of the task for which it was used, e.g., classification vs. generation:

- **Classification model validation metrics:** The most frequently applied approach (14 papers) to evaluate requirements quality is adapted from ML and uses metrics used to evaluate supervised classification models. Studies use metrics such as *F1-score*, *Precision*, *Recall*, and *Accuracy* for quality assessment models that provide categorical-scale outcomes. This includes studies using SVMs, kNNs, CNNs, and transformer-based classifiers (e.g., RoBERTa, BERT, SBERT). These metrics were especially prevalent in the detection of Ambiguity and Clarity.
- **Expert evaluations and agreements with experts:** In 7 studies, expert-based evaluations were conducted to assess the quality, relevance, or usefulness of AI-generated outputs, particularly in cases where no objective ground truth was available. These evaluations were typically embedded in user-centered or exploratory setups. For instance, Paper [46] employed *expert satisfaction ratings* to assess the effectiveness of LLM-based agent systems in interpreting and evaluating user stories. The setup involved six agile teams interacting with AI agents during requirements clarification and backlog grooming sessions, providing detailed feedback on the agents' performance and usefulness. Similarly, Paper [62] used the *\$100 prioritization technique* to evaluate how effectively LLMs could assist in prioritizing user stories. In Paper [65], experts evaluated the efficiency of AI models in extracting semantically relevant frequent nouns from textual requirements. In addition to such subjective assessments, a subset of three papers [50,61,64] applied quantitative agreement-based metrics between expert and AI assessments, such as the percentage of requirements for which both parties reached the same quality judgment. These evaluations were especially relevant for quality aspects such as Usability and Estimability.
- **Task-specific or contextual metrics:** The three papers [37,50], and [63] employed customized, domain-specific evaluation metrics such as *defect detection rate* or *consistency of quality scores* over multiple assessment repetitions. These metrics were typically applied in scenarios that closely reflect real-world or practice-oriented use cases. For example, Paper [63] assessed how effectively LLMs could detect defects in requirement documents by directly measuring the number of identified quality-related issues, using this as a proxy for practical utility. Similarly, Paper [50] used ChatGPT to evaluate the quality of user stories in a structured setting that closely resembles actual industry practices, with the aim of validating its potential as a semi-automated quality reviewer. Such task-specific metrics were particularly employed to assess nuanced quality aspects such as Negotiability and Independency, where standard classification metrics were insufficient or inapplicable.
- **Similarity and generation metrics:** In the two Papers [34,53], metrics such as *Bilingual Evaluation Understudy (BLEU)*, *BERTScore*, *AlignScore*, and *Semantic Entropy Score (SE-Score)* were used to evaluate quality assessment methods involving generative tasks or semantic transformation of requirements. These metrics evaluate how closely the AI output resembles a reference (often human-produced) one. They are mostly associated with instruction-tuned transformers and used for Ambiguity detection and Clarity evaluation.

5.4.3. Evolution AI-based quality assessment methods over time

A longitudinal view of the 26 analyzed papers reveals distinct temporal patterns in the adoption of AI techniques for assessing requirements quality. The methods show a clear evolution from traditional ML and NLP approaches toward more sophisticated and large-scale generative models. Specifically, we observed the following trends:

- **2019–2021: Classical ML and NLP techniques dominate the landscape.**

Earlier studies in this period (e.g., [47,66]) predominantly employed classical NLP features and ML classifiers as well as CNNs. These approaches often required manual feature engineering or predefined linguistic metrics to identify requirements quality issues such as ambiguity or vagueness. Evaluation was typically based on popular metrics such as F1-score, Precision, and Recall.

- **2022–2023: Transformer-Based models gain in traction.**

From 2022 onward, the adoption of transformer-based models such as BERT, RoBERTa, and SBERT increases markedly (e.g., Paper [48,65]). These models enable deeper semantic understanding of requirements artifacts and reduce the need for prior engineering of relevant features. These approaches are typically evaluated using the same classification metrics as classical ML models but often outperform the latter in precision and robustness.

- **2023–2025: The raise of LLMs.**

Starting in late 2023 and accelerating into 2024 and 2025, a significant shift toward LLMs such as GPT, LLaMA, Codex, and Bloom becomes evident. These models are frequently used not just for classification but also for generation, summarization, and automated critique of requirements (e.g., Paper [37,62,63]). This trend co-occurs with a diversification of evaluation strategies, including BLEU, BERTScore, and subjective expert evaluations, and reflects a movement toward human-like understanding and evaluation of requirements quality.

- **2024–2025: Hybrid and multi-agent approaches mark a recent trend.**

Another recent trend involves hybrid approaches that combine LLMs with rule-based or multi-agent frameworks, such as in Paper [46]. These setups aim to enhance explainability, robustness, and coverage of the generated quality assessments by leveraging the strengths of multiple models or agents.

- **Evaluation metrics evolve in parallel.**

Evaluation strategies follow the methodological shift in quality assessment approaches. While classical ML models rely on quantitative metrics like F1 or accuracy, LLM-based approaches often include semantic similarity (e.g., BLEU, AlignScore) or expert-based qualitative judgments. This suggests a methodological change of the field, recognizing that approaches based on GenAI require fundamentally different validation strategies than classical classification models.

6. Discussion

6.1. What dimensions of requirements quality are considered?

Requirements quality assessment methods consider multiple quality aspects, which encompass various categories and abstraction levels. Despite this variability, many quality aspects used in the related literature exhibit semantic and functional overlap. To address this fragmentation and increase practical applicability, we mapped the identified aspects to the INVEST model proposed by Wake [45]. This model is widely adopted in agile RE and provides a sparse yet expressive framework for evaluating user stories based on six core dimensions: *Independent*, *Negotiable*, *Valuable*, *Estimable*, *Small*, and *Testable*.

The mapping shown in Fig. 5 demonstrates that each of the quality aspects identified in our study can be mapped to one of the six INVEST criteria:

- **Independent (I):** Includes *Independency* and *Atomicity*, both of which reflect the need for modular, self-contained user stories.
- **Negotiable (N):** Covers *Negotiability*, *Ambiguity*, *Vagueness*, and *Clarity*, highlighting the interpretative flexibility and communicative clarity needed during collaborative refinement.
- **Valuable (V):** Encompasses *Usability* and *Appropriateness*, ensuring that the user story delivers relevant value to stakeholders.

Table 12
Overview of evaluation metrics and corresponding studies.

Evaluation metric	Studies
F1-Score	[49], [47], [48], [50], [55], [56], [57], [35]
F2-Score	[49]
Recall	[49], [68], [47], [48], [50], [55], [57], [58], [51], [35]
Precision	[49], [68], [47], [48], [50], [55], [57], [51], [35]
Accuracy	[54], [57], [58], [60], [35]
Specificity	[50]
Absolute count of true positives	[59]
Agreement between model and benchmark (e.g., percentage)	[61], [64], [50]
Expert's subjective evaluation of model's efficiency	[65]
Expert's overall satisfaction score	[46]
Prioritization evaluation (e.g., \$100 method)	[62]
Mean deviation between model and expert assessments	[52]
Consistency over multiple runs	[50]
Defect Detection Rate	[63]
Average Satisfaction Score	[37]
BLEU Score	[53]
SE Score (Semantic Entropy)	[53]
BERTScore	[34]
AlignScore	[34]
None reported	[61], [67]

- **Estimable (E):** Groups *Estimability*, *Feasibility*, and *Consistency*, focusing on the predictability and planning reliability of requirements.
- **Small (S):** Relates to *Conciseness*, *Structure*, *Completeness*, and *Complexity*, all of which influence the manageable size and clarity of a user story.
- **Testable (T):** Includes *Testability*, *Correctness*, *Redundancy*, and *Traceability*, reflecting the verifiability and validation potential of requirements.

This mapping shows that although quality aspects are diverse in terminology and granularity, they can be meaningfully reduced to a standardized framework. The INVEST model offers such a high-level classification structure that can be refined further depending on domain-specific needs. Such alignment not only supports consistent evaluation but also enhances practical usability in agile contexts. At the same time, some of the more granular aspects, such as *Traceability* or *Consistency*, may still be relevant for more formal or regulated development environments and should not be disregarded.

Finally, the evolution of quality assessment methods, specifically toward GenAI, coincides with a diversification of requirement quality aspects. While earlier works focused primarily on *Completeness* and *Ambiguity*, more recent papers address more subjective aspects such as *Understandability*.

6.2. For what purpose and in which RE phase is the quality of requirements assessed?

The mapping of quality aspects to the purpose and phase of their application reveals both dominant trends and notable underexplored areas. As expected, the *Characterize/Evaluate* purpose dominates in the studies we analyzed. This underscores the diagnostic character of current research, where most quality models focus on recognizing quality problems rather than solving them, e.g., by prescribing the ways of resolving quality issues or by directly generating improved requirements.

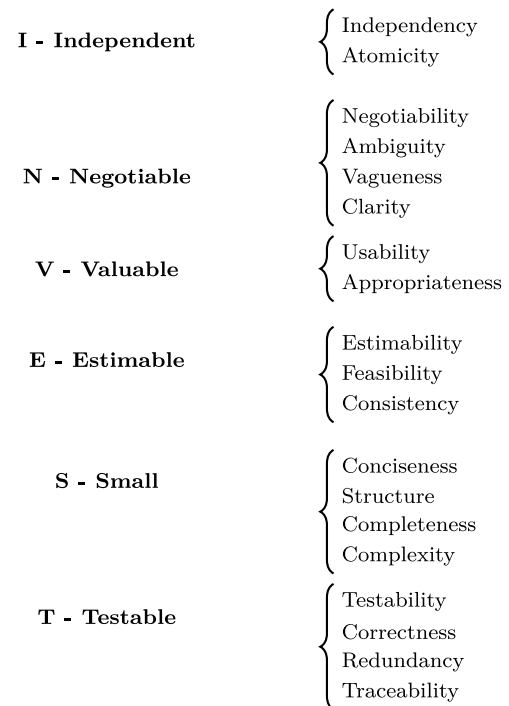


Fig. 5. Mapping of Quality Aspects to INVEST Criteria. Diagram illustrating how different quality aspects of user stories relate to the INVEST principles (Independent, Negotiable, Valuable, Estimable, Small, Testable) using braces to group associated sub-aspects.

The high frequency of the studies where quality assessment aimed at *Explain/Understand* and *Improve* purposes indicates an increasing interest not only in assessing but also in comprehending quality deficits,

including their possible root causes, consequences, and solution potentials. Quality aspects such as *Testability*, *Clarity*, and *Conciseness* are frequently considered in the context of both understanding and improvement purposes, reflecting their relevance in iterative requirements refinement. In contrast, the *Predict* purpose remains marginal, assigned to only four quality aspects and pointing to the limited role of predictive modeling in current requirements quality assessment frameworks.

In terms of RE lifecycle integration, the *Analysis* phase is most frequently addressed. This reflects a continued reliance on post-elicitation review practices and suggests that quality assurance activities often take place once requirements have been initially captured but are still documented in relatively unstructured or semi-structured formats. With 15 quality aspects associated, the *Elicitation* phase also plays a central role, especially for linguistic and stakeholder-facing dimensions, directly affecting the comprehensibility and usability of requirements for non-technical stakeholders, such as *Testability*, *Conciseness*, or *Negotiability* during requirements discussion. This emphasis underlines the importance of capturing quality early, when user needs are still being verbalized.

Later stages are less frequently addressed. For example, the fact that the quality aspects *Ambiguity*, *Correctness*, *Usability* are explicitly considered in the *Modeling* phase, may indicate that quality is rarely assessed once requirements take structured form. The complete absence of the *Management* phase in the context of analyzed studies further illustrates a gap in RE lifecycle-spanning quality governance.

Taken together, these findings suggest that current quality frameworks remain predominantly evaluative and centered on early- to mid-stage requirements activities. To fully support agile and iterative development, future work should aim to bridge quality assessment into the late engineering phases of the RE lifecycle and to the management phase, and develop tools that offer guidance for improvement, not just analysis.

6.3. What metrics are used to quantify the quality of requirements?

The predominance of *binary metrics* (Yes/No) highlights a clear tendency toward minimal effort and scalable evaluations of quality aspects in user stories. While effective for identifying discrete issues such as the presence of ambiguity or non-compliance with INVEST principles of Wake [45], such binary classifications may oversimplify nuanced aspects like *Clarity* or *Testability*. These limitations call for more graded or context-sensitive measures.

Likert-scale approaches offer a more flexible evaluation, capturing degrees of quality through subjective judgment. However, their reliability is contingent on evaluator expertise and consistency, which can vary significantly across application contexts. Interestingly, only a handful of papers applied *Likert* ratings in a structured and repeatable way, limiting their generalizability.

Novel contributions emerged with the adoption of LLMs, which specifically targets the semantic aspects of requirements quality. However, these contributions are rare, indicating potential for future development and a lack of standardization in advanced evaluation techniques.

Finally, the heterogeneity of metrics and their applications points to a fragmentation in how quality is conceptualized and assessed across the literature. While some overlap exists, particularly among linguistic quality aspects, many studies seem to invent ad hoc metrics tailored to their specific research setup. Having a standardized framework or metric taxonomy for assessing requirement quality would likely facilitate better comparability and cumulative knowledge building in this area.

6.4. What data is used to develop and validate AI-based methods for requirements quality assessment?

Similar to quality aspects, metrics, evaluation methods, and datasets used in the context of AI-based requirements quality assessment suggest

high fragmentation and limited standardization of this field. While relatively many datasets were considered in the reviewed studies (35 datasets in 26 papers), a different dataset is used in almost every study; only four of the datasets, *PURE*, *PROMISE*, *ReqEval* and *DAMIR*, appear in multiple studies.

The overall comparability of methods proposed by such studies is limited by the largely heterogeneous structure, labeling, and scope of the used datasets. For example, while some datasets contain thousands of user stories or requirement sentences (e.g., *PURE*, *DAMIR*), others are limited to a handful of specification documents or pages. Heterogeneity of data affects not only model training strategies but also the types of quality aspects that can be meaningfully analyzed. Fine-grained sentence-based datasets support linguistic metrics such as *Vagueness* or *Clarity*, whereas document-level datasets are more suitable for structural or traceability-focused analysis.

The limited availability of proprietary datasets additionally limits the comparability and replicability of the associated quality assessment methods and models. Almost half of the datasets are proprietary and the prevalence of synthetic data further complicate the transfer of the methods, models, and the associated study findings to real-world settings. While synthetic data allow for controlled experimentation and scalability, they often lack the ambiguity, inconsistency, and domain-specific terminology that characterize genuine requirements artifacts. The large ratio of 26 labeled datasets out of 35 in total surprised us positively, particularly if we consider that almost all of them (25) were labeled by human experts, which implies a high labor cost of preparing such data. Still, preparing labeled data in the context of requirements quality assessment remains one of the major concerns, especially regarding scalability. Due to its high cost, manual labeling is limited to small datasets.

6.5. What AI methods and models are adapted for assessing the quality of requirements?

Our analysis reveals a clear evolution in AI-based approaches for assessing software requirements quality, highlighting a progressive shift from traditional to advanced methods. Initially, classical ML techniques such as *SVM*, *kNN*, and *Naïve Bayes* were widely employed, relying on handcrafted features extracted through conventional NLP preprocessing. These approaches were primarily effective for detecting structural and surface-level quality issues such as *Structure* and *Vagueness*, favored for their interpretability and computational efficiency.

With the advent and adoption of transformer-based models, notably *BERT*, *RoBERTa*, and *SBERT*, the field advanced toward richer semantic understanding of requirements text. These models generate dense contextual embeddings that better capture nuanced linguistic and semantic patterns, enabling improved detection of abstract quality concerns such as *Clarity*, *Ambiguity*, and *Consistency*. This transformer-centric preprocessing, often combined with downstream classifiers, reduced the need for manual feature engineering and enhanced generalization capabilities.

The most prominent methodological innovation is the widespread integration of LLMs such as *GPT-3.5*, *GPT-4*, *LLaMA* variants, and related architectures. These models excel in tasks involving open-ended, context-dependent quality dimensions like *Ambiguity*, *Conciseness*, and *Testability*, where their generative and reasoning capabilities enable sophisticated rewriting, critique generation, and evaluative judgments. The LLM-centric pipelines often minimize explicit preprocessing, leveraging the models' pretrained contextual knowledge to lower the barrier for quality assessment.

Additionally, hybrid pipelines that combine LLM-based preprocessing (e.g., *BERT* embeddings or *GPT* vectorization) with classical ML classifiers represent an effective middle ground, harnessing the rich semantic representations from transformers while maintaining the efficiency and interpretability of classical algorithms. These mixed approaches, though less common, demonstrate promising results in ambiguity detection and other quality dimensions.

Less frequently, DL models like *CNNs* have been applied to capture local and sequential textual patterns, showing potential for detecting qualities such as *Completeness*, but their adoption remains limited due to data and complexity demands.

Emerging research directions include multi-agent and hybrid architectures that integrate LLMs with symbolic reasoning or structured knowledge components, aiming to balance the flexibility and contextual prowess of LLMs with explainability and control. Although still nascent, these approaches may represent the future trajectory for robust, interpretable AI-assisted requirements quality assessment.

A final observation concerns the degree of model adaptation. Most studies employing transformer-based models such as BERT or SBERT fine-tuned them or repurposed pretrained embeddings for RE-specific datasets, indicating domain adaptation through representation learning. In contrast, recent work increasingly relies on off-the-shelf, general-purpose GenAI systems such as ChatGPT or GPT-4, typically without fine-tuning, using prompt engineering or multi-agent configurations instead. This shift highlights a methodological divide between domain-adapted language models and general-purpose generative models, with the former emphasizing controlled task performance and the latter offering flexibility and broader semantic coverage. Future research may benefit from systematically comparing these two strategies regarding effectiveness, explainability, and reproducibility in RE practice.

In summary, the RE-community is increasingly leveraging the semantic richness and generative power of LLMs for complex quality dimensions, while transformer-based embeddings and classical ML methods continue to provide complementary strengths in efficiency, interpretability, and handling structural quality attributes.

6.6. How the performance of AI-based quality assessment methods is evaluated?

The evaluation of AI-based quality assessment methods reveals methodological pluralism, yet also inconsistencies and gaps. On the one hand, widespread use of standard classification metrics, such as *Precision*, *Recall*, *Accuracy*, or *F1-score*, enables a degree of comparability across studies. On the other hand, these metrics are insufficient for evaluating assessment tasks other than classification or semantic aspects of AI-based quality assessment, such as semantic alignment of AI-based and human-based quality assessments. Yet, with the methodological advancements, especially the adoption of GenAI, suitable evaluation methods and metrics are increasingly used, for example, semantical similarity-based *BLEU* or *BERTScore* metrics. Since these often conflate lexical overlap with functional adequacy, another promising direction is incorporating expert judgment, either via direct rating or structured methods such as the *\$100 Prioritization Technique*. However, this approach is far from being perfect; its inherent weakness of these methods is their high subjectivity, especially since they are rarely combined with quantitative metrics.

A further concern is the limited attention to inter-rater agreement and reproducibility. Only a few papers report reliability measures or provide open datasets. This hinders replicability and undermines confidence in findings, especially given the mentioned inherent subjectivity of human judgments.

6.7. How quality assessment methods exploit advancements in AI technologies?

A temporal analysis of the AI-based requirements quality assessment reveals a clear alignment of their evolution and advancement in AI technologies. Although with a certain lag, the RE-community consequently adapts recent advancements in AI to improve the capabilities of requirements quality assessment methods. Before 2022, quality assessment predominantly used classical supervised ML methods based on performance tightly coupled to domain-specific structured datasets. From 2023 onward, there is a quite rapid uptake of GenAI techniques

and prompting strategies. This is not only a technical trend but reflects a broader shift in moving from simply detecting and classifying quality issues to generating issue explanations, improvement suggestions, or directly the improved requirements.

7. Threats to validity

While this SLR was conducted with a rigorous methodology, several threats to validity should be acknowledged.

Selection bias. The search process may have introduced bias toward certain types of publications. Despite using multiple databases and snowballing techniques, some studies may have been omitted due to limited keyword coverage or publication in non-indexed venues. This could particularly affect the representativeness of industrial approaches or non-English publications.

Construct validity. Our categorization of quality aspects and their mapping to broader frameworks such as INVEST is based on interpretive synthesis, which involves a degree of subjectivity. Although we applied peer discussions, there remains a risk that nuances or contextual definitions of quality aspects in individual studies were simplified. The thematic grouping may also reflect a normative bias toward agile evaluation models.

Internal validity. Some studies lacked detailed descriptions of their datasets, annotation procedures, or model architectures. This lack of transparency may have led to misclassification or misinterpretation of methods or results.

External validity. Many of the reviewed models rely on synthetic or proprietary datasets that may not generalize to real-world RE scenarios. The limited reuse of benchmark datasets across studies further influences generalizability. Moreover, since only a few papers explicitly addressed integration into RE workflows, the practical applicability of these methods remains uncertain.

Publication bias. There is a risk that studies reporting successful applications of AI in RE are more likely to be published than those with negative or inconclusive results. This potential bias may distort the perceived effectiveness of AI-based quality assessment methods.

8. Research opportunities and challenges

The analysis revealed several research gaps that define opportunities and challenges for advancing AI-based quality assessment of software requirements.

A major challenge lies in the lack of a unified terminology and taxonomy for defining quality aspects. The reviewed studies employ heterogeneous terminologies, and overlapping or similar concepts of quality concepts using different terms. Developing a standardized taxonomy of quality aspects, building on the established INVEST framework, for which we showed, a mapping of seemingly different quality aspects exists, would improve comparability of results.

Another methodological limitation concerns the inconsistency and lack of standardization in the evaluation metrics used across studies. The reviewed literature applies a wide range of quantitative and qualitative metrics. However, these metrics differ in granularity, which makes comparison of the results across studies difficult. This variability highlights the need for a consolidated framework of metrics that aligns with the different purposes (e.g., characterization, evaluation, improvement) and phases (e.g., analysis, management) of requirements quality assessment.

Another methodological challenge concerns the reproducibility of AI-based results. Current studies often rely on small, export-labeled, or synthetic datasets, and few provide replication packages due to proprietary ownership. Establishing high-quality, open, and annotated

benchmark datasets for requirements quality assessment would greatly improve the transparency and transferability of the field.

Although recent work increasingly adopts LLMs and generative AI, only a few studies explore their full potential to go beyond diagnostic tasks. Most studies use models to detect or classify requirements based on predefined quality aspects. Just a few explore the capabilities of novel GenAI models to understand discovered issues and obtain improvement suggestions. Research could further investigate how AI systems can interpret discovered issues and generate actionable suggestions for refinement. This requires progress in explainable and traceable AI models that can justify their decisions and recommendations within RE contexts.

Additionally, hybrid architectures that combine symbolic reasoning with generative models offer a promising direction. Such systems could bridge the gap between deep semantic understanding and explicit, rule-based reasoning about requirements quality, leading to more transparent and trustworthy assessment tools.

From an application perspective, transferring AI-based quality assessment methods into industrial RE practice remains difficult. The diversity of models and the technical expertise required to configure and maintain them hinder adoption. Opportunities lie in designing adaptive, user-friendly toolchains that integrate with existing RE workflows and support practitioners without requiring deep AI expertise.

Finally, the ethical and governance aspects of AI-assisted quality assessment pose emerging challenges. Ensuring fairness, accountability, and data privacy in models trained on sensitive project artifacts will be crucial for their acceptance and long-term sustainability.

9. Directions for future research

Building upon the identified gaps and challenges, the following research directions are suggested to advance the field:

- **Development of Benchmark Datasets:** Establish publicly available, high-quality datasets with standardized annotation schemes for evaluating AI-based requirements quality assessment methods. These datasets should capture diverse requirement types and domains to improve generalizability.
- **Standardization of Quality Criteria and Evaluation Metrics:** Define a common taxonomy and set of metrics for measuring requirements quality, allowing the consistent evaluation and comparison across studies.
- **Developing Reusable and Explainable Models:** Develop interpretable AI systems capable of explaining the rationale behind quality assessments and improvement suggestions, improving trust and facilitating adoption in practice.
- **Integration with RE Processes:** Investigate how AI-based assessment can be embedded into agile and continuous RE environments, supporting real-time feedback and iterative quality management.
- **Improvement-Oriented and Hybrid Approaches:** Extend existing descriptive models to other purposes of requirements analysis and management, including generating constructive improvement suggestions, specifically with hybrid AI architectures that combine symbolic reasoning with generative capabilities.
- **Empirical Validation in Industrial Settings:** Conduct longitudinal studies in industrial environments to evaluate the practical benefits, scalability, and maintainability of AI-based quality assessment systems.

Following this agenda will create an environment in which recent advancements in AI can more efficiently be adapted and RE practitioners can more effectively and reliably adapt AI-based quality assessment approaches in their specific application contexts.

10. Conclusions

This review systematically synthesized the state of the art on applying AI techniques to assess the quality of software requirements. By analyzing 26 primary studies published between 2019 and 2025, the review identified the quality aspects considered, the datasets and metrics used, and the predominant AI methods utilized to develop and evaluate AI-based quality assessment approaches.

The findings indicate a rapidly evolving research landscape characterized by a methodological shift from traditional ML and NLP toward LLMs and multi-agent architecture. These models enable semantically rich analysis of textual requirements and support new levels of automation in quality assessment. At the same time, the field remains fragmented in terminology, dataset usage, and evaluation practices, which hinders replicability and the comparability of results across studies. Despite growing methodological sophistication, the practical integration of AI-based quality assessment into RE processes remains limited.

Overall, the field can be characterized as emerging, dynamic, and yet lacking methodological standardization. The synthesis provided in this review forms a structured foundation for future work and can support researchers and practitioners in navigating this evolving area.

CRedit authorship contribution statement

Elise Wolf: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Conceptualization. **Adam Trendowicz:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Conceptualization. **Julien Siebert:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

It is publicly available here:

[Empirical Data Package for “Quality Assessment of Software Requirements Using Artificial Intelligence Methods: A Systematic Literature Review” \(Original data\) \(Mendeley Data\)](#)

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